

FULL ENVELOPE AIRSPEED CALIBRATION USING A SINGLE TURNING ACCELERATION TECHNIQUE

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1. INTRODUCTION AND BACKGROUND

Virtually every performance or flying qualities flight test program begins with the calibration of either the aircraft's production Pitot-static system and/or a dedicated Air Data Boom (ADB). One of the main goals in Pitot-static calibration is to determine the Static Source Position Error (SSPE), or the error induced in Pitot-static instrument readings due to the disturbance of the local static pressure field by the aircraft itself [1]. SSPE is typically manifested in errors in Indicated Airspeed (IAS) and altitude, both of which rely on accurate static pressure readings. Although there are currently many Flight Test Techniques (FTTs) designed to collect Pitot-static calibration data, they are all based on the general principle of flying the aircraft at known conditions such as Ground Speed (GS) and altitude, while simultaneously comparing the known conditions to instrument readings.

Altitude calibration techniques pose significantly less challenges due to the ease and simplicity associated with flying the aircraft at a known geometric and pressure altitude using the prolific Tower Flyby FTT. In contrast, airspeed calibration techniques present additional challenges due to the common inability to measure (and differentiate between) IAS, GS, and wind speed during test execution. Additionally, since the static field disturbances that generate SSPE are produced by the act of flying the aircraft, the airspeed and altitude errors found during calibration tend to be functions of airspeed itself [1]. Therefore, in order to characterize SSPE in both airspeed and altitude, a range of test points covering a reasonable portion of the aircraft's flying envelope must be flown.

This paper aims to provide a robust framework for the data collection, reduction, and processing of airspeed calibration data as well as a novel FTT for obtaining such data. The combination of the proposed data framework and method for obtaining data will be shown to drastically improve the test efficiency and data quality associated with the airspeed calibration of any flight test program. Prior to identifying the underlying gaps in the state of the art methods, it is important to briefly compare the existing methods.

SOCIETY OF FLIGHT TEST ENGINEERS

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1.1 Current Methods

As previously mentioned, most current FTTs focus on flying the aircraft at known conditions and comparing the known parameters to instrument readings. Additionally, airspeed calibration poses a more challenging problem when compared to altitude calibration due to the inability to directly measure Calibrated Airspeed (CAS), especially in the presence of wind. Up to date, the challenge of measuring CAS has been solved by either flying at or near Sea Level (which allows for the approximation of CAS using True Airspeed (TAS) or GS and wind), or calculating CAS using TAS, altitude, and indicated temperature. As such, the following sections focus on how some of the current FTTs solve the challenge of measuring CAS.

1.1.1 Ground Speed Course

In [1], Erb describes the classical Ground Speed Course, which prescribes flying a two-leg, reciprocal heading, straight line course that maintains airspeed and heading while allowing the aircraft ground track to drift with wind. The purpose of the reciprocal headings is to “average out” wind speed using the assumption that wind speed and direction remain constant during test execution. This allows for the approximation of CAS given an average computed GS. The technique is then repeated for a series of IASs within the aircraft’s flying envelope.

1.1.2 Three-legged GPS Course

In [2], National Test Pilot School (NTPS) improves on the Ground Speed Course by taking advantage of Global Positioning System (GPS) as a truth source for GS and Ground Track. The Three-Legged GPS FTT prescribes three orthogonal straight-line legs where airspeed and heading are tightly held. Once flown, basic trigonometric formulas are used to determine wind speed and direction, which allow the computation of TAS and CAS. Again, this technique is repeated for every airspeed of interest in the aircraft’s flying envelope.

1.1.3 Cloverleaf and Level Turn

A subsequent and logical improvement to the previously listed FTTs involves taking advantage of the high-speed data recording capabilities now found in most flight test programs, as well as linear regression or statistical computation methods. As such, [3] and [4] provide the Cloverleaf and Level Turn FTTs, respectively. Both FTTs prescribe flight paths where the aircraft’s heading is varied to provide a linear regression matrix with adequate measurement observability to estimate CAS, wind speed, and wind direction. These FTTs not only allow for but prescribe variations in heading during test execution, which makes test execution less difficult. However, IAS must still be tightly held since the techniques must be repeated for each desired airspeed in the aircraft’s flying envelope. Finally, the use of regression algorithms expands the possibilities for data quality feedback by allowing the computation of confidence intervals on the results.

1.2 Research Contributions

The Cloverleaf and Level Turn FTTs are able to collect the necessary data to differentiate TAS (and subsequently CAS) from wind speed by providing the regression matrix with measurement observability in heading. That is, by changing heading during the execution of the test point, the regression matrix is able to accurately decompose the measured CAS into actual CAS and wind speed components.

The research provided in this paper expands the concept of measurement observability into the airspeed domain. Using the Level Turn FTT and associated regression matrices described by Jorris in [4], a novel FTT referred to from hereupon as the Turning Acceleration FTT, can be developed. In the Turning Acceleration FTT, heading and airspeed are uniformly changed throughout the test point by accelerating the aircraft while executing the turn. The result is an airspeed calibration technique that not only tolerates, but prescribes changes in heading and airspeed throughout the test point. More importantly, the new technique is able to produce the entire SSPE curve for an aircraft using only a single test point.

The accompanying regression algorithm explained further in later sections utilizes Akaike Information Criterion (AIC) as described in [5] and [6] to decide on the most appropriate SSPE curve shape for each aircraft from one of many pre-programmed family of shapes. Additionally, it provides a confidence interval around the curve itself, which can be used to determine data quality and algorithm confidence. Finally, a Monte-Carlo experiment focusing on the effects of wind variability is provided for three example aircraft to give the user an accurate tradeoff analysis between wind variability and expected confidence interval size.

2. METHODOLOGY

The problem that all airspeed calibration methods seek to solve is to define SSPE (ΔV), which is simply the error between IAS (V_{IAS}) and CAS (V_{CAS}) in the airspeed case. ΔV can then be used to correct V_{IAS} and provide the pilot with an accurate estimate of V_{CAS} using the relationship

$$V_{CAS} = V_{IAS} + \Delta V. \quad (1)$$

In general, ΔV varies with V_{CAS} . As V_{CAS} itself is not measurable, the challenge in airspeed calibration is to design an experiment in which V_{CAS} can be inferred from other measurable data. Observations of ΔV over a range of V_{CAS} are then used to build an approximate model of ΔV as a function of either V_{CAS} or V_{IAS} . This model then becomes the source of ΔV in practical applications of (1). The next task is to define the airspeed error function.

2.1 Computing the SSPE Curve

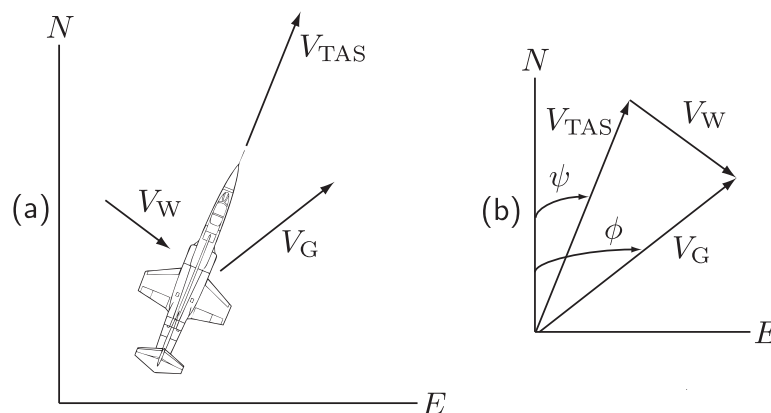


Figure 1: Reference System Definition.

The focus of this subsection is to develop the machinery required to compute a SSPE curve

from a very generalized set of test data, assuming that these test data are available. It provides a common framework within which various techniques for collecting the flight test data may be compared on equal terms and allows for the computation of SSPE curves with fewer assumptions than traditional data reduction techniques. The mechanics of actually collecting the flight test data are discussed in subsection 3, but before the flight test techniques are discussed, it helps to define the data requirements and data reduction process.

2.1.1 Basic Relationships and Definitions

It is useful to begin by providing relationships between the physical quantities that describe the aircraft's planar motion. Let the TAS vector be $V_{TAS} = (V_{TAS_E}, V_{TAS_N})^T$, the wind vector be $V_W = (V_{W_E}, V_{W_N})^T$, and the GS vector be $V_G = (V_{G_E}, V_{G_N})^T$, with components defined along the east and north axes. Then

$$V_{TAS} + V_W = V_G \quad (2)$$

describes the relationship between the three vectors, as seen in Figure 1. Note that TAS, rather than IAS or CAS, is required if one desires to form the mathematical relationship described in (2). However, thus far, only the relationship between IAS and CAS has been defined, in (1). One can construct a very simple relationship between these quantities as follows. First, from [1], the explicit definitions of CAS and TAS are

$$V_{CAS} = a_0 \sqrt{5 \left[\left(\frac{q_c}{P_0} + 1 \right)^{\frac{2}{7}} - 1 \right]} \quad (3)$$

$$V_{TAS} = a_0 \sqrt{5 \frac{T}{T_0} \left[\left(\frac{q_c}{P} + 1 \right)^{\frac{2}{7}} - 1 \right]} \quad (4)$$

where a_0 is the speed of sound at sea level on a standard day, T is the static air temperature, T_0 is the temperature at sea level on a standard day, q_c is the impact pressure, P_0 is the pressure at sea level on a standard day, and P is the static pressure. Note that when $T = T_0$ and $P = P_0$, $V_{CAS} = V_{TAS}$. Then, by restricting oneself to collecting data at or near ambient conditions (P_0, T_0) , one may simply interchange CAS and TAS. Relaxing this assumption is certainly possible, but it requires the use of (4) for the remainder of the discussion. For simplicity, it is assumed that the test is conducted at (P_0, T_0) for the remainder of this discussion.

Given the $V_{CAS} = V_{TAS}$ assumption, and a definition of the IAS vector as $V_{IAS} = (V_{IAS_E}, V_{IAS_N})^T$, substituting (1) into (2) yields an expression relating indicated airspeed to ground speed,

$$V_{IAS} + \Delta V + V_W = V_G. \quad (5)$$

In practice, it is difficult to measure the components of IAS and GS; instead, magnitude and direction are typically measured. The direction of both the TAS and IAS vectors relative to the north axis is the aircraft heading ψ , while the direction of the GS vector relative to the north axis is the aircraft course ϕ . Also, in general, the wind vector V_W is not measurable.

Given these limitations, and test measurements for V_{IAS} and V_G , the objective is to build a model $\mathcal{M}_\Delta$ of ΔV as a function of V_{IAS} so the aircraft's IAS may be corrected to CAS. Typically, $\mathcal{M}_\Delta$ will

Model Name	Mathematical Form	Type
Polynomial(N)	$\mathcal{M}_\Delta(V_{IAS}; \mathbf{w}) = \sum_{i=0}^P w_i V_{IAS}^i$	Linear
Power	$\mathcal{M}_\Delta(V_{IAS}; \mathbf{w}) = w_0 + w_1 V_{IAS}^{w_2}$	Nonlinear
Exponential	$\mathcal{M}_\Delta(V_{IAS}; \mathbf{w}) = w_0 + w_1 \exp(w_2 V_{IAS})$	Nonlinear

Table 1: Example model families used in fitting SSPE curves.

be parameterized by a set of P weights $\mathbf{w} = (w_1, \dots, w_p)^\top$, which are inferred from the test data. Some examples of suitable $\mathcal{M}_\Delta$ are shown in Table 1.

2.1.2 Fitting the SSPE Model Curve

To begin, note that

$$\Delta V = \mathcal{M}_\Delta(V_{IAS}; \mathbf{w}) + R(V_{IAS}), \quad (6)$$

where $R(V_{IAS})$ is the residual (model error) in $\mathcal{M}_\Delta$ at V_{IAS} . Equation (6) can be rearranged as

$$R = \mathcal{M}_\Delta(V_{IAS}; \mathbf{w}) - \Delta V, \quad (7)$$

which can then be minimized in a least-squares sense over a set of N test data to compute an estimate of $\mathbf{w}$ as

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_{\forall n \in N} R_{(n)}^2 = \arg \min_{\mathbf{w}} \sum_{\forall n \in N} \left[\mathcal{M}_\Delta(V_{IAS(n)}; \mathbf{w}) - \Delta V_{(n)} \right]^2. \quad (8)$$

Many algorithms are capable of solving (8); the MATLAB routine `fitnlm` was chosen for this particular application. Typically, such algorithms desire a set of known data $\mathbf{y}$, a feature matrix Φ consisting of the model inputs where each column is a unique feature and each row is a unique observation, and a handle to a function that maps Φ and a set of weights $\mathbf{w}$ to the outputs (typically written $(\Phi, \mathbf{w}) \mapsto \mathcal{M}_\Delta(\Phi; \mathbf{w})$). It would then use an iterative method, such as the Levenberg-Marquardt method, to converge on an estimate for $\hat{\mathbf{w}}$ that solves (8).

One can rearrange equation (5) to suit this purpose. Assuming S test data samples are available and taking V_{IAS} , V_G , ψ , and ϕ as knowns, ΔV and V_W as unknowns, and splitting the vectors into their north and east components, eq. (5) can be rearranged as

$$V_G - V_{IAS} = V_W + \Delta V, \quad (9)$$

$$\begin{bmatrix} V_{G_N} - V_{IAS_N} \\ V_{G_E} - V_{IAS_E} \end{bmatrix} = \begin{bmatrix} V_{W_N} + \Delta V_N \\ V_{W_E} + \Delta V_E \end{bmatrix}, \quad (10)$$

$$\underbrace{\begin{bmatrix} V_{G_{(1)}} \cos \phi_{(1)} - V_{IAS_{(1)}} \cos \psi_{(1)} \\ \vdots \\ V_{G_{(S)}} \cos \phi_{(S)} - V_{IAS_{(S)}} \cos \psi_{(S)} \\ V_{G_{(1)}} \sin \phi_{(1)} - V_{IAS_{(1)}} \sin \psi_{(1)} \\ \vdots \\ V_{G_{(S)}} \sin \phi_{(S)} - V_{IAS_{(S)}} \sin \psi_{(S)} \end{bmatrix}}_{\stackrel{\text{def}}{=} \mathbf{y} \in \mathbb{R}^{2S}} = \begin{bmatrix} V_{W_N} \\ \vdots \\ V_{W_N} \\ V_{W_E} \\ \vdots \\ V_{W_E} \end{bmatrix} + \underbrace{\begin{bmatrix} \Delta V_{(1)} \cos \psi_{(1)} \\ \vdots \\ \Delta V_{(S)} \cos \psi_{(S)} \\ \Delta V_{(1)} \sin \psi_{(1)} \\ \vdots \\ \Delta V_{(S)} \sin \psi_{(S)} \end{bmatrix}}_{\approx \mathcal{M}_\Delta(\Phi; \mathbf{w}) \in \mathbb{R}^{2S}}, \quad (11)$$

$$\mathbf{y} \approx \begin{bmatrix} V_{W_N} \\ \vdots \\ V_{W_N} \\ V_{W_E} \\ \vdots \\ V_{W_E} \end{bmatrix} + \mathcal{M}_\Delta(\Phi; \mathbf{w}). \quad (12)$$

Equation (12) holds for all reasonable choices of $\mathcal{M}_\Delta$, including those shown in Table 1. Note the definitions of $\mathbf{y}$ and $\mathcal{M}_\Delta(\Phi; \mathbf{w})$. Also, note that if wind effects are not negated by the flight test technique used to obtain the test data, values for the components of V_W will also have to be obtained when fitting $\hat{\mathbf{w}}$. This may be accomplished by simply replacing the right hand side of eq. (12) with

$$\mathbf{y} \approx \begin{bmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_{W_N} \\ V_{W_E} \end{bmatrix} + \mathcal{M}_\Delta(\Phi; \mathbf{w}) \quad (13)$$

and taking V_{W_N} and V_{W_E} as free parameters when fitting the model. All that remains is to define Φ before the reader is fully equipped to fit a model to test data. Unlike $\mathbf{y}$, Φ depends on the particular model chosen. For example, the Polynomial(N) model has

$$\begin{aligned}
\Phi = \text{diag} & \left(\begin{bmatrix} \cos \psi_{(1)} \\ \vdots \\ \cos \psi_{(S)} \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right) \begin{bmatrix} 1 & V_{IAS(i)}^1 & \cdots & V_{IAS(i)}^P \\ \vdots & \vdots & & \vdots \\ 1 & V_{IAS(S)}^1 & \cdots & V_{IAS(S)}^P \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \\
+ \text{diag} & \left(\begin{bmatrix} 0 \\ \vdots \\ 0 \\ \sin \psi_{(1)} \\ \vdots \\ \sin \psi_{(S)} \end{bmatrix} \right) \begin{bmatrix} 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \\ 1 & V_{IAS(i)}^1 & \cdots & V_{IAS(i)}^P \\ \vdots & \vdots & & \vdots \\ 1 & V_{IAS(S)}^1 & \cdots & V_{IAS(S)}^P \end{bmatrix}. \quad (14)
\end{aligned}$$

Since the polynomial model is linear, that is $\mathcal{M}_{\Delta}(\Phi; w) = \Phi w$, in the no-wind case, equation (13) can be rewritten as

$$y \approx \Phi w \quad (15)$$

and solved¹ in a least-squares sense as

$$w = (\Phi^T \Phi)^{-1} \Phi^T y. \quad (16)$$

In the nonlinear case, as for the exponential and power families of models, an iterative method is required, so for the sake of generality, the routine called “FITNLM” will be referenced when solving for w . One could repeat this process and arrive at a set of weights for several model families. The question is then which model or models to choose to best represent the SSPE curve.

2.1.3 Choosing the Best Model

Having fit several models to the same data, the next task becomes choosing which model best describes the process that generated the data. Many techniques are well-suited for this task. The technique favored by the authors and employed in this investigation is to use the Akaike Information Criterion (AIC),

$$AIC = 2p - 2 \log \mathcal{L}, \quad (17)$$

where p is the number of parameters (i.e. coefficients) in the model, and $\mathcal{L}$ is the maximized value of the model's likelihood function. In practice, AIC must be corrected for a finite sample size, and this corrected quantity is referred to as Corrected Akaike Information Criterion (AICc). AICc is defined as

¹Note that the system is overdetermined, so the task is to choose the “best” solution.

$$AICc = AIC + \frac{2p(p+1)}{S-p-1} \quad (18)$$

where S is the sample size.

AIC and AICc have their roots in information theory, and interested readers are directed to [5] for the theoretical background. Briefly, AIC and AICc are measures of how much information is lost in replacing the generating process (i.e. truth or reality) with a model. It balances how well the model fits the data, represented by the log likelihood $\log \mathcal{L}$, with the complexity of the model, represented by the number of parameters p . Clearly, if forced to choose a single model from a finite set of models fit to a data set, one should choose that model with the least information loss and with reasonable complexity. However, AIC and AICc provide an additional benefit, in that they can be used to combine models. When it is sensible to do so, mixing models allows for less information to be lost than choosing only one model. To determine what is “sensible”, one must compute the relative Akaike weights π_k of the candidate models.

Let $\delta_k = AICc_k - \min\{AICc_1, \dots, AICc_K\}$ be the relative AICc for each of the K models. Then the Akaike weight of each model is given by

$$\pi_k = \frac{\exp\left(-\frac{1}{2}\delta_k\right)}{\sum_{\kappa=1}^K \exp\left(-\frac{1}{2}\delta_{\kappa}\right)}, \quad (19)$$

and by comparing the π_k one can determine which model or models best describe the test data. Note that this formulation of π_k ensures that the model weights are normalized (sum to 1). Thus the combined model may be represented as a linear combination of the individual models,

$$\mathcal{M}_{\Delta}(V_{IAS}; \{\mathbf{w}_k, \pi_k\}_{\forall k \in K}) = \sum_{\kappa=1}^K \pi_{\kappa} \mathcal{M}_{\Delta}(V_{IAS}; \mathbf{w}_{\kappa}). \quad (20)$$

This is critical because it allows the models to be fit independently of one another and combined after the fact.

In practice, some of the π_k may be small, indicating that the corresponding model is not very informative. It is useful to discard the non-informative models and recompute the π_k excluding those models. Doing so is a simple matter of reusing equation (19) with the δ_k corresponding to the offending models excluded.

2.1.4 Implementing the Method

Having now thoroughly discussed the technical background of the model, it is important to return to our original goal of characterizing the SSPE curve for a particular aircraft. At this point, the machinery required to compute the SSPE curve from test data is available. All that remains is to collect the flight test data required to fit the models. A myriad of choices are available, and in the following section, a few existing methods are compared.

2.2 Classical Methods for Obtaining Flight Test Data

From the previous section, it is apparent that GS magnitude V_G , IAS magnitude V_{IAS} , heading ψ and ground track ϕ need to be collected during the flight test if one hopes to compute a SSPE

curve. Also, wind effects must either be negated by the flight test technique, measured and incorporated into the model, or inferred from the test data. As mentioned before, it is difficult to measure the wind speed and direction reliably enough to arrive at passable results for the SSPE curve, so most techniques either focus on averaging out wind effects or ensuring that wind effects are observable enough to allow for inference of the wind speed and direction in post-processing. Also, if a high degree of accuracy is desired, then many useable samples must be taken during the flight test, either by completing a large number of test points or by leveraging the high sample rate of modern data acquisition equipment.

A brief discussion of three well-known techniques follows to motivate the technique proposed by this paper.

2.2.1 Ground Speed Course

The ground speed course method, as described in reference [1], uses a series of N test points, each conducted at a constant² IAS $\bar{V}_{IAS(n)}$, to form point estimates of ΔV at each $\bar{V}_{IAS(n)}$. Note that the $\bar{V}_{CAS(n)}$ cannot be measured directly; however, it can be inferred from the average GS $\bar{V}_{G(n)}$ and IAS $\bar{V}_{IAS(n)}$ from each run. Since wind effects are eliminated if the FTT is executed properly, from equation (2), $\bar{V}_{CAS(n)}$ is equal to $\bar{V}_{G(n)}$ ³. The weights w for $\mathcal{M}_{\Delta}(V_{IAS}; w)$ are then fit using the point estimates at the N airspeeds $\bar{V}_{IAS(n)}$ as training data.

The primary drawbacks of the ground speed course method are that for the assumptions to hold, tight tolerances on airspeed and altitude must be held throughout the maneuver. Also, a single test point yields only a point estimate of ΔV at the prescribed V_{IAS} . As N is typically of order 10 due to practical constraints, parameter estimates $\hat{w}$ formed typically have wide confidence intervals, yielding imprecise predictions of ΔV . Note that the data collected with this technique fit within the bounds of the data reduction framework proposed in the previous section. Concatenating the results of all N test points into a single vector and modifying equation (12), yields

$$\begin{bmatrix} V_{G(1)} - V_{IAS(1)} \\ \vdots \\ V_{G(N)} - V_{IAS(N)} \end{bmatrix} = \begin{bmatrix} \Delta V_{(1)} \\ \vdots \\ \Delta V_{(N)} \end{bmatrix} \approx \mathcal{M}_{\Delta}(\Phi; w), \quad (21)$$

where w can be fit for an arbitrary number of models $\mathcal{M}_{\Delta}$ as described above.

2.2.2 Level Turn and Cloverleaf Methods

The level turn method, as described in reference [4], and the cloverleaf method, as described in [3], are evolutions of the ground speed course method (and many other FTTs and are designed to leverage the high sample rates of modern instrumentation, data acquisition systems, and computational methods to obtain more precise predictions of ΔV . Both methods also use a series of N test points, each conducted at an average IAS $\bar{V}_{IAS(n)}$, but for each test point, M measurements of $V_{IAS(m,n)}$ and $V_{G(m,n)}$ are taken. Typically M is of order 10^3 or more; consequently, for each of the N test points, a probability distribution of $\Delta V_{(n)}$ is obtained rather than a simple point estimate. Additionally, wind effects may be incorporated into the model as an additional latent variable, so a

²Assumed constant.

³This depends critically on the assumptions made in the previous subsection.

probability distribution of $V_{W(n)}$ is also obtained for each of the N test points; also, the models are tolerant of wind noise. The weights w for $\mathcal{M}_\Delta(V_{IAS}; w)$ are then fit using all available data at the N airspeeds, yielding more precise parameter estimates than were afforded by the ground speed course method.

Clearly, these methods are similar to the ground speed course. The primary structural differences are the addition of wind as a latent variable and the incorporation of many (M) realizations of the observed variables for each of the N test points. Thus, stronger estimates of the latent variables may be formed for the same number of test points N by using the level turn or cloverleaf methods rather than the ground speed course method. Note that the data collected with these and similar techniques also fit within the bounds of the data reduction framework proposed in the previous section. Again, taking ΔV to be constant for each test point, and modifying equation (12), yields

$$\begin{bmatrix} V_{G(1)} \cos \phi_{(1)} - V_{IAS(1)} \cos \psi_{(1)} \\ \vdots \\ V_{G(M)} \cos \phi_{(M)} - V_{IAS(M)} \cos \psi_{(M)} \\ \vdots \\ V_{G(1)} \sin \phi_{(1)} - V_{IAS(1)} \sin \psi_{(1)} \\ \vdots \\ V_{G(M)} \sin \phi_{(M)} - V_{IAS(M)} \sin \psi_{(M)} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cos \psi_{(1)} \\ \vdots & \vdots & \vdots \\ 1 & 0 & \cos \psi_{(M)} \\ 0 & 1 & \sin \psi_{(1)} \\ \vdots & \vdots & \vdots \\ 0 & 1 & \sin \psi_{(M)} \end{bmatrix} \begin{bmatrix} V_{WN} \\ V_{WE} \\ \Delta V_n \end{bmatrix} \quad (22)$$

to compute ΔV (and the wind, which is discarded) at the prescribed $\bar{V}_{IAS(n)}$ for each of the N test points. Then, by concatenating the results from the N test points into a single vector, one may simply use

$$\begin{bmatrix} \Delta V_{(1)} \\ \vdots \\ \Delta V_{(N)} \end{bmatrix} = \mathcal{M}_\Delta(\Phi; w) \quad (23)$$

to fit the SSPE curve. Again, this is an improvement over the ground speed course method because it generates higher fidelity estimates of each $\Delta V_{(n)}$, ultimately yielding a higher-fidelity estimate of the SSPE curve.

While methods of this type offer many benefits and makes better use of modern instrumentation, data acquisition equipment, and computational techniques than the ground speed course (and many other methods), there are still opportunities to streamline the method to further test efficiency.

2.3 A New Technique for Obtaining the Data

The focus of this paper is the turning acceleration method. The turning acceleration method uses a single test point conducted over the full operational range of V_{IAS} of the aircraft to fit a curve to ΔV over the entire range of V_{IAS} . During the test point, M measurements of V_{IAS} , V_G , ϕ and ψ are collected using high-speed data acquisition equipment (M is of order 10^3 or greater). Additionally, since the aircraft is maneuvered through a full turn, wind effects are observable and are incorporated into the model as an additional latent variable in the same manner as the cloverleaf and level turn methods. The model remains tolerant of wind noise. The key difference between this method and the previous method is that no assumptions is made that the test point is conducted at a fixed

V_{IAS} (yielding a fixed ΔV), rather, V_{IAS} is assumed to vary during the test point. This yields many benefits, including increased tolerance of wind noise and execution error. Data reduction requires the full model described in (12) above since ΔV is no longer assumed to be constant throughout the test point. However, the entire SSPE curve may be calculated in one step, and from one test point.

3. FLIGHT TEST TECHNIQUE

Although the proposed algorithm and FTT were developed in support of a C-130J flight test program, the concepts, mathematics, and flight maneuvers necessary to perform the calibration algorithm apply to all aircraft. In general, the flight maneuver is as simple as flying the aircraft on a constant bank, coordinated 360 degree accelerating turn while maintaining altitude. It is important to note that sideslip directly introduces error into the algorithm since it corrupts the trigonometric relation between the heading (ψ), ground track (ϕ), and wind direction (φ_W) angles. The remaining details needed to successfully fly the technique are discussed below.

3.1 Altitude Selection

All airspeed calibration techniques rely on the adequate comparison of the TAS, GS, and wind vectors. As such, all airspeed calibration techniques, including this one, are ideally flown as close as possible to sea level (where Knots Indicated Airspeed (KIAS) and TAS are approximately equal) or at higher altitudes with additional computational and instrumentation requirements in order to calculate TAS from KIAS. Furthermore, the altitude tolerance required during the turn should be based on the desired confidence interval on the final correction curve. Using a general rule of thumb derived from cursory model sensitivity analysis, the maneuver tolerance should be ± 50 feet in order to assure a confidence interval no greater than ± 0.5 knots.

3.2 Angle of Bank Selection

Since Angle of Bank (AOB) directly drives turn radius, its selection may inevitably be driven by airspace constraints, especially with high speed aircraft. However, there are key considerations that must be addressed when implementing the proposed FTT for a particular aircraft. In general, most airspeed calibrations are conducted in order to generate configuration-dependent airspeed correction curves for aircraft flying at wings level. Since the error in airspeed measurement is generated, in large, by a disturbance of the static field around the aircraft, the factors affecting the disturbance characteristics should be carefully examined. Specifically, the static field disturbance has been shown to be a function of airspeed (and by corollary, a function of Angle of Attack (AOA) and load factor) [1]. Therefore, it is important to balance the airspace restrictions with the changes in AOA and load factor as a functions of airspeed, which are induced by changes in AOB. As such, it is recommended that the proposed FTT be flown at the lowest practical AOB in order to maintain the AOA-airspeed relation as close as possible to that of a wings-level aircraft. The mathematical algorithms described in this paper will yield a valid solution regardless of the angle of bank selected. However, the resulting correction curve may not necessarily match the curve of a wings-level aircraft if AOB selection significantly increased the AOA for a given airspeed.

3.3 Acceleration Profile

Similar to the method used in basic algebra to solve a system of linear equations, the statistical regression algorithm used in the proposed FTT (as well as the Level Turn and Cloverleaf) works

by converging on a solution (a set of unknowns) given enough valid measurements (independent equations). The quality of the measurements provided to the algorithm directly affects the confidence interval of the final solution. By executing at least one full turn, the algorithm is given enough independent measurements to converge on the wind speed and direction unknowns. Similarly, by providing a wide range of airspeeds, the algorithm can then converge on the airspeed correction unknowns. As a result, it is recommended that the proposed FTT be flown using the widest possible range of airspeeds for a given gear and flap configuration profile. It is important to note that extrapolation beyond the data bounds of a particular model is seldom valid or accurate. Therefore, the airspeed range used during the execution of the proposed FTT should also take into consideration the calibration curve requirements driving the test (i.e. the range of airspeeds desired for the final calibration curve). Furthermore, in order to minimize the effects of wind variability on the measurement collection, the acceleration profile should remain as constant as possible. However, if airspace restrictions and aircraft-specific operating limitations yield an airspeed profile that is fully accelerated prior to completing a full turn, it is recommended that the remainder of the turn be completed at the last obtained airspeed. Finally, the current version of the algorithm is unable to model piecewise functions such as the ones created by transitions into transonic and supersonic regions. Therefore, a separate turning acceleration test point should be performed for each major airspeed region of the desired correction curve.

3.4 Instrumentation Data Rate

Although modern instrumentation systems are capable of recording at high data rates, it is important to carefully investigate the relationship between the recording system's sampling rate and the physical properties of the variables being recorded. That is, the rate of change of recorded variables should be analyzed and matched to the number of unique (independent) samples available at such rate. If any particular variable of interest changes at a slower rate than the number of samples recorded per second, the resulting repeated measurements allowed into the algorithm will falsely improve the Confidence interval (CI) of the results. In this particular case, the authors noticed the available data rate (100 Hz) far surpassed the Pitot-static system's ability to detect and provide independent samples. As such, prior to running the algorithm, the recorded data were filtered (yielding an effective sampling rate of 16 Hz) in order to remove repeated values that would otherwise artificially improve the confidence interval of the results. As later shown in Figure 3, the CI resulting from the filtered data was 0.88 kts while the unfiltered data yielded a CI of 0.35 kts.

4. RESULTS AND CONCLUSIONS

4.1 Numerical Results

In order to determine the usefulness of the turning acceleration method as an FTT, numerical simulations were conducted for the Lockheed C-130J aircraft using flight manual SSPE curves as a truth source. An acceleration profile in CAS was prescribed, and the flight manual SSPE curve was then used to generate the IAS profile that would be seen in the aircraft and recorded by the data acquisition system. In order to demonstrate the algorithm's stability against variable or gusty wind, a mean wind component with uniformly distributed additive noise was added to the airspeed data. Finally, the variable wind was added to the CAS track in order to create the measured GPS track for the aircraft. The simulated data for V_{IAS} , V_G , ϕ , and ψ , was processed using the methods described in this paper in an attempt to recreate the truth data V_{CAS} and the SSPE curve (ΔV) obtained from the aircraft's flight manual.

Figure 2 shows the estimated velocity correction curve with confidence intervals overlaid on the flight manual velocity correction curve. All candidate models were fit (quadratic, cubic, power, and exponential), and a mixture model was selected, indicating that none of the models had an Akaike weight $pi > 0.7$ and that a mixture of several models provided a better fit than a single model for this particular realization of additive wind noise. The goodness of fit is similar to the C172 curve (note the y -scale of the graph).

A Monte-Carlo experiment was then conducted on the effects of increasing wind variability, and the results are shown on the right side of Figure 2. This chart can be used to estimate an acceptable level of gust intensity for a specified tolerance for the velocity correction curve confidence intervals; for example, if one desires a confidence interval of 1 knot, test wind gust limits should be set to approximately 4 knots.

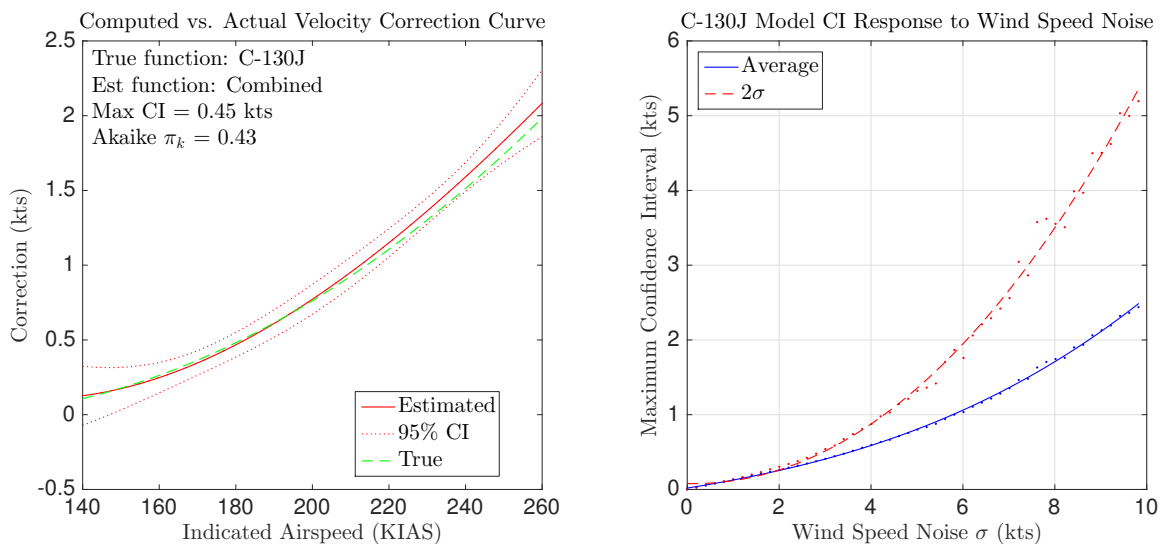


Figure 2: C-130J Numerical Results

4.2 Flight Test Results

In order to validate the proposed algorithm, the proposed FTT was flown in along with the ground speed course and the level turn to calibrate a wingtip air data boom. As shown in Figure 3, the SSPE curve obtained using a single Turning Acceleration test point falls well within the confidence intervals of the curves obtained by the other two methods, each requiring 4 test points. Table 2 summarizes the flight time savings of the Turning Acceleration and Level Turn FTTs compared to the Ground Speed Course.

	FTT	Test Points	Flight Time	Savings
Ground Speed Course		4	36.5 min.	-
Level Turn		4	15.7 min.	57%
Turning Acceleration		1	4.6 min.	87%

Table 2: Flight test time comparison.

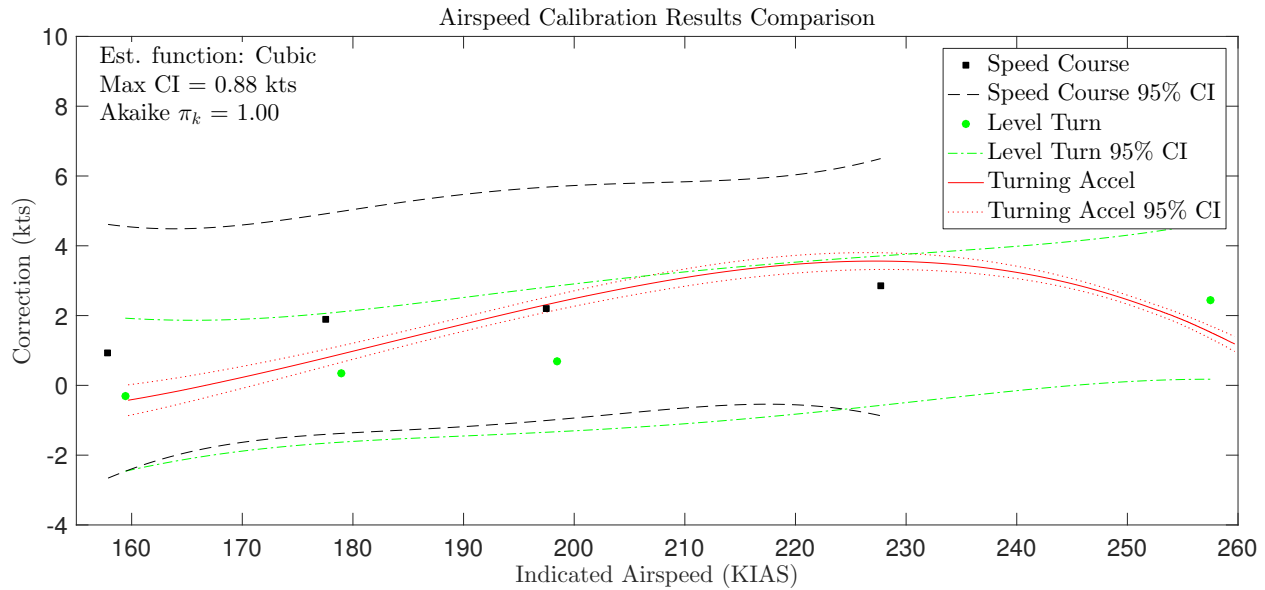


Figure 3: Flight Test Results.

4.3 Conclusions

As is evident from Table 2, the Turning Acceleration method offers significant time savings when compared to other existing methods. In fact, a sufficiently sophisticated data acquisition system could allow an air data system be calibrated in flight if it used a real-time implementation of the method in only a few minutes. This could allow a test team to recalibrate their Pitot-static system every flight, if so desired.

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Captain Clark C. McGehee is a Flight Test Engineer student at the USAF Test Pilot School. Prior to his selection into TPS, he was a C-130J Flight Test Engineer at the 413th Flight Test Squadron, Hurlburt Field, FL where he was responsible for planning, executing, and reporting on various C-130J test programs. Captain McGehee is a 2010 graduate of Duke University. After graduation, he remained at Duke University and earned a Master of Science and Doctorate of Philosophy in Mechanical Engineering.